

GII Index Reduction Analysis Report

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1 Executive Summary

This report documents a method for determining how far the number of indicators in the Global Innovation Index (GII) 2025 can be reduced without significantly compromising the index’s discriminatory power. The method was previously applied to the EU JRC’s Sustainable and Inclusive Well-being Index (Benczur et al. 2024; Boskovic et al. 2024); this report replicates it on GII’s 78 Level-1 indicators observed for 139 countries/economies.

Two perspectives are evaluated: a **country perspective** (do countries end up the same distance apart using a subset as with the full indicator set?) and a **variable perspective** (does a subset preserve the PCA structure of the full set?). These use two different metrics that are easy to conflate but measure different things: the country perspective reports a **Spearman rank correlation** between full- and subset-distance matrices; the variable perspective reports r_m (and r_m^2 , interpretable as a % of the full PCA’s variance structure recovered by the subset).

Per the project’s specified methodology - one indicator per sub-pillar, chosen as the sub-pillar’s best- r_m “reflective indicator” - the headline recommendation is a **21-indicator subset** ($r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full PCA’s variance structure; Rand Index 0.902 against the full-data country clustering). A larger 42-indicator subset, built by allocating indicators proportionally to each pillar’s actual size rather than a flat count per pillar, reaches $r_m^2 = 92.6\%$ and a rank correlation of 0.941 with the full country-distance matrix. Both are documented below alongside the flat and proportional strategies they were chosen from.

At the 21-indicator budget, exhaustive within-group search does not clearly outperform an unconstrained random subset of similar size, for either metric; the gap becomes more visible at the larger 42-indicator budget. Both perspectives’ results below discuss this directly.

2 Introduction

When creating indicator dashboards and composite indices, indicator selection often involves narrowing down from an initially larger set. Besides data quality and relevance considerations, statistical methods are essential to assess whether indicators within a well-defined component are sufficiently aligned. A key statistical question is: how much of the information content (“variation”) of the full indicator set is preserved in a candidate subset?

This analysis applies two methodological approaches to the GII 2025 data (78 Level-1 indicators, 139 countries):

1. **Country Perspective:** Assessing how similar countries are using all indicators versus a subset, using distance matrices and clustering techniques
2. **Variable Perspective:** Using Principal Component Analysis (PCA) to measure how much information a subset captures compared to the full set

2.1 Finding a suitable reduced set

Composite indices such as the GII combine two different measurement logics. At the upper levels - pillars, sub-indices, the index itself - the structure is *formative*: each pillar captures a distinct dimension of innovation performance, and the index’s meaning depends on all of them being represented; dropping a pillar entirely would change what the index measures, not just how precisely it measures it. Within a sub-pillar, by contrast, the indicators are closer to *reflective* measures of a single underlying concept - largely interchangeable manifestations of the same thing, where one or two well-chosen indicators can stand in for the group without changing what is being measured.

This distinction motivates the reduction strategy used throughout this analysis: every pillar and sub-pillar is retained in every candidate subset, preserving the formative structure and therefore the index’s meaning, while indicator reduction happens only *within* each group, by selecting which reflective indicator(s) best represent it. It also motivates why two different allocation rules are tested later (flat vs. proportional, see Methodology): both only decide *how many* reflective indicators to draw from each formative group, never *whether* to include the group at all - with one exception discussed in Methodology, where proportional allocation can assign a very small sub-pillar zero indicators.

Even restricted to searching *within* a single pillar, exhaustive search grows quickly: selecting the best possible subset of size k out of p indicators by brute force requires evaluating $C(p, k) = \frac{p!}{k!(p-k)!}$ combinations. The Institutions pillar (IN.1) has 6 indicators, so choosing 2 of them already means 15 possible combinations. Multiplied across all 7 pillars, the number of combinations for a full-index subset is far beyond exhaustive search - which is why this analysis searches exhaustively within each pillar or sub-pillar (a tractable sub-problem) rather than

across the whole index at once. The table below shows how many combinations remain even after allocating a 20-indicator budget proportionally across pillars.

Table 1: Unique combinations per pillar of GII when allocating a 20-indicator budget proportionally

Pillar	Indicators in Pillar	Allocated (budget=20)	Combina- tions
Institutions	6	1	6
Human capital and research	12	3	220
Infrastructure	9	2	36
Market sophistication	11	3	165
Business sophistication	15	4	1365
Knowledge and technology outputs	14	4	1001
Creative outputs	11	3	165

Even after this proportional allocation collapses the problem to one pillar at a time, there are still 1,767,710,124,180,000 combinations to explore in total - underscoring why within-group exhaustive search (rather than search across the full index) is the only tractable approach.

3 Methodology

3.1 Statistical Methods Explanation

To determine how much of the information content of the full indicator set is preserved in a potential subset, the variable perspective uses a subset-selection approach based on Principal Component Analysis (PCA). Rather than simply ranking individual variables by importance, this method evaluates entire subsets of variables according to how well they reproduce the PCA structure of the full dataset.

This criterion, r_m , was introduced by McCabe (1984) as one of several optimality criteria for selecting “principal variables” - subsets of the original variables that play the role principal components play in ordinary PCA, but remain individually interpretable because they are actual measured indicators rather than linear combinations of all of them. Cadima and Jolliffe (2001) formalize this same criterion as one of two indicators for how well a subset of variables approximates a full PCA (the other being a subspace-similarity measure, the generalized coefficient of determination), and show that r_m^2 is exactly the percentage of total variance in the original data accounted for by regressing the full data on the chosen subset - the interpretation used throughout this report. Both papers show, with worked examples, that choosing variables by their PC loadings alone - a common shortcut - frequently produces subsets that are clearly

suboptimal, and sometimes badly so, compared to subsets chosen by directly maximizing r_m ; this is part of the motivation for using exhaustive (rather than loadings-based) search within each pillar and sub-pillar throughout this analysis.

A key advantage of this method is its built-in protection against redundancy. For example, if multiple indicators are highly correlated (nearly identical), including one in the subset provides most of the information, and adding the others contributes relatively little additional ability to reconstruct the principal components.

The mathematical formula for the selection criterion is:

$$r_m = \sqrt{\frac{\sum_{i=1}^p \lambda_i r_{mi}^2}{\sum_{j=1}^p \lambda_j}}$$

where: - λ_i is the eigenvalue of principal component i from the full dataset (representing the variance explained by that component) - m indexes a specific candidate subset - r_{mi}^2 is the coefficient of determination (R^2) between principal component i and the candidate subset of variables. Since PC_i is a linear combination of the original variables, the R^2 of the subset measures how well the principal component can be reconstructed from the subset.

Conceptually, this formula computes a weighted average of how well the subset predicts each principal component, with weights given by the variance explained by each component, then normalizes by the total variance. The resulting r_m^2 value can be interpreted as the percentage of total variation explained by the subset, providing an intuitive measure of information retention.

This method assumes that “information” is defined by the variance structure captured by PCA, which is appropriate for statistical data reduction. However, it may downweight variables with low variance but high policy relevance, which is an important consideration for substantive index development (see (Algursan et al. 2026) for a discussion of the broader limitations of variance-based, GDP-adjacent metrics for capturing societal progress).

Evaluating r_m against the *full* 78-indicator PCA requires regressing every one of the 78 principal components on the candidate subset - expensive enough (one `lm()` fit per component per subset) that it is only used to score a small number of *assembled* candidate subsets, never as the search criterion for exhaustive search itself (see Proportional Allocation Methodology, below).

3.1.1 The Country Perspective's Distance-Based Criterion

The variable perspective's r_m criterion asks whether a subset preserves the *variance structure* of the indicators. The country perspective asks a more direct question: does a subset preserve how similar or dissimilar countries are judged to be relative to one another? For each candidate subset, the pairwise Euclidean distance between every pair of countries is computed using only that subset's indicators, and the ranking of those pairwise distances is compared to the ranking

obtained from the full indicator set, via Spearman rank correlation. A correlation near 1 means the subset reproduces almost the same country-similarity ordering as the full index; a low correlation means the subset materially reshuffles which countries look similar to which. Unlike r_m , this criterion makes no reference to principal components or variance explained - it is a direct comparison of two similarity structures. The same distance-preservation logic, applied to group structure rather than pairwise ranks, underlies the SIMPROF clustering comparison used later in this report (see Clustering Results).

3.2 Proportional Allocation Methodology

Both perspectives select candidate subsets by exhaustive search *within* each pillar or sub-pillar, then assemble the per-group picks into a full subset - consistent with the formative/reflective distinction introduced in “Finding a suitable reduced set” above: every pillar and sub-pillar is represented in every candidate subset, and only the reflective indicators within each group are subject to selection. This requires deciding, up front, how many indicators to take from each group. Two allocation rules are used throughout this report:

- **Flat allocation** (Strategies 1 and 2): every pillar contributes the same fixed number of indicators (3 for Strategy 1), or every sub-pillar contributes 1 (Strategy 2). This is simple, but ignores that GII’s pillars range from 6 indicators (Institutions) to 15 (Business sophistication), and its sub-pillars range from 2 to 5 - a flat rule systematically over-represents small groups and under-represents large ones relative to their actual share of the index.
- **Proportional allocation** (Strategies 1a and 2a): each group’s budget is set proportional to its share of all indicators, using a **largest-remainder (Hamilton) apportionment**: every group first receives `floor(share)` indicators, and any indicators left over from rounding are handed out one at a time to the groups with the largest fractional remainder. This is the same method used to apportion legislative seats to regions by population. Unlike flat allocation, a group with a very small share of the index can receive **zero** indicators under this rule - the one case, noted above, where a formative group is not represented in the resulting subset. The total budget used is **42** indicators (double Strategies 1/2’s 21): with 21 sub-pillars, a 21-indicator budget would leave no room for proportionality to show up at the sub-pillar level (`floor` alone already allocates exactly 1 to every sub-pillar), so a larger budget is needed for Strategy 2a to differ meaningfully from Strategy 2.

Within each allocated group (of either kind), the actual indicators are chosen by exhaustive search maximizing a *local* r_m (for the variable perspective) or a local proxy computed the same way (for the country perspective) - restricted to that group’s own sub-PCA rather than the full 78-indicator PCA. This local search is a tractable stand-in for an otherwise infeasible full search; the assembled subset is then re-scored against the true, full-dataset metric that is actually reported in this document.

3.3 Data Description

The Global Innovation Index (GII) 2025 dataset ((WIPO) 2024) (`gii-data/idata.xlsx`, with metadata in `gii-data/imeta.xlsx`) comprises 78 Level-1 indicators observed for 139 countries/economies, organized in a 5-level hierarchy: 78 indicators -> 21 sub-pillars -> 7 pillars -> 2 sub-indices (Inputs, Outputs) -> 1 overall index. Indicators flagged with `Direction == -1` in the metadata (e.g. pupil-teacher ratio, applied tariff rate) are reversed ($\max(x) - x$) before standardization, so that higher standardized values consistently indicate better innovation performance across all indicators. Missing values are imputed via k-nearest-neighbors (VIM: :kNN, $k = 5$) prior to z-scoring; see the Appendix for imputation statistics.

4 Key Findings

4.1 Country Perspective

4.1.1 Distance Matrix Analysis: Random Subsets vs. Full Dataset (Fig 2)

For subset sizes k ranging from 10% to 90% of all 78 indicators (plus $k = 42$, matching the proportional strategies' budget below), 1,000 random subsets of each size were drawn and compared to the full-dataset Euclidean distance matrix via Spearman rank correlation:

Table 2: Rank correlation of random subsets vs. the full distance matrix, by subset size k

Subset Size (k)	Mean Correlation	Median Correlation	Min Correlation	Max Correlation
8	0.678	0.687	0.383	0.824
15	0.784	0.789	0.626	0.878
23	0.847	0.851	0.726	0.918
31	0.891	0.893	0.808	0.936
38	0.920	0.922	0.856	0.955
42	0.932	0.935	0.872	0.962
46	0.946	0.948	0.898	0.971
54	0.963	0.965	0.902	0.983
62	0.978	0.979	0.945	0.991
69	0.988	0.989	0.957	0.996

Finding: Correlation rises smoothly and near-monotonically with subset size, from a mean of 0.678 at $k = 8$ to 0.988 at $k = 69$. This random-subset curve is the baseline the exhaustive-search strategies below are judged against.

4.1.2 Exhaustive Search: Strategies 1 and 2 (Flat Allocation)

Table 3: Strategy 1: best 3 indicators per pillar (flat), by rank correlation with the full distance matrix

Pillar	r_m of Best Subset	Selected Indicators
Institutions	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	0.757	Expenditure on education; Gross expenditure on R&D; Tertiary enrolment
Infrastructure	0.784	Gross capital formation; ICT access*; Low-carbon energy use
Market sophistication	0.764	Domestic credit to private sector; Domestic market scale; Joint venture deals
Business sophistication	0.746	Females employed w/advanced degrees; High-tech imports; University–industry R&D collaboration†
Knowledge and technology outputs	0.702	Software spending; Production and export complexity; Patents by origin
Creative outputs	0.697	Entertainment and media market; National feature films; Industrial designs by origin

Table 4: Strategy 2: best 1 indicator per sub-pillar (flat), by rank correlation with the full distance matrix

Sub-Pillar	r_m of Best Subset	Selected Indicators
Institutional environment	0.938	Operational stability for businesses*
Regulatory environment	0.975	Rule of law*
Business environment	0.834	Entrepreneurship policies and culture†
Education	0.626	School life expectancy
Tertiary education	0.615	Tertiary enrolment
Research and development (R&D)	0.793	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.877	ICT access*
General infrastructure	0.647	Logistics performance*
Ecological sustainability	0.588	ISO 14001 environmental certificates

Sub-Pillar	r_m of Best Subset	Selected Indicators
Credit	0.650	Finance for startups and scale-ups†
Investment	0.750	Joint venture deals
Trade, diversification and market scale	0.639	Domestic industry diversification
Knowledge workers	0.802	Knowledge-intensive employment
Innovation linkages	0.727	University–industry R&D collaboration†
Knowledge absorption	0.508	Research talent
Knowledge creation	0.638	PCT patents by inventor origin
Knowledge impact	0.592	High-tech manufacturing
Knowledge diffusion	0.649	Production and export complexity
Intangible assets	0.613	Industrial designs by origin
Creative goods and services	0.550	Entertainment and media market
Online creativity	0.639	GitHub commits

Finding: Strategy 1 (21 indicators, 3/pillar) reaches a correlation of 0.834; Strategy 2 (21 indicators, 1/sub-pillar) reaches 0.841. Both are close to - not clearly above - the random-subset baseline at a comparable size (mean 0.847 at $k = 23$ in the table above), suggesting that at a 21-indicator budget, exhaustive within-group search buys little over chance for this metric.

4.1.3 Clustering Results: SIMPROF + Rand Index

Countries were clustered via SIMPROF (significance-tested hierarchical clustering, see Clarke et al. (2008)) on the full 78-indicator dataset and separately on the better-scoring of Strategies 1/2 (Strategy 2, per its higher correlation above), then compared via Rand Index:

Table 5: Rand Index: full-data clustering vs. Strategy 2 clustering

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Strategy
Full_vs_Sub-set	0.902	78	21	Strategy2_1perSub-pillar

Finding: The 21-indicator subset recovers a Rand Index of 0.902 against the full-data clustering - a substantially closer match than the rank-correlation metric alone would suggest, indicating that the coarse cluster structure is more robust to indicator reduction than fine-grained distances are.

4.1.4 Proportional Allocation: Strategies 1a and 2a

Strategies 1 and 2 above allocate a *flat* number of indicators to every pillar or sub-pillar. Strategies 1a and 2a instead allocate a 42-indicator budget proportionally, using the largest-remainder method described in Methodology above.

Table 6: Strategy 1a: proportional allocation across pillars

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutions	3	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	6	0.898	Expenditure on education; Global corporate R&D investors; Researchers; Graduates in science and engineering; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	5	0.900	GDP/unit of energy use; Gross capital formation; ICT access*; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.926	Applied tariff rate, weighted avg.; Domestic industry diversification; Joint venture deals; Late VC deals intensity; Market capitalization; Loans from microfinance institutions
Business sophistication	8	0.932	Emerging knowledge workforce; FDI net inflows; GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Patent families; State of cluster development†
Knowledge and technology outputs	8	0.915	Software spending; Labor productivity growth; High-tech exports; ICT services exports; ISO 9001 quality certificates; PCT patents by inventor origin; Unicorn valuation; Utility models by origin
Creative outputs	6	0.865	Mobile app creation; Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; National feature films; Trademarks by origin

Table 7: Strategy 2a: proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutional environment	1	0.938	Operational stability for businesses*
Regulatory environment	1	0.975	Rule of law*
Business environment	1	0.834	Entrepreneurship policies and culture†
Education	3	0.882	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.840	Graduates in science and engineering; Tertiary inbound mobility
Research and development (R&D)	2	0.922	Global corporate R&D investors; Researchers
Information and communication technologies (ICTs)	2	0.966	Government's online service; <i>ICT use</i>
General infrastructure	2	0.867	Electricity output; Gross capital formation
Ecological sustainability	1	0.588	ISO 14001 environmental certificates
Credit	1	0.650	Finance for startups and scale-ups†
Investment	3	0.980	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.639	Domestic industry diversification
Knowledge workers	3	0.944	Females employed w/advanced degrees; GERD financed by business; GERD performed by business
Innovation linkages	3	0.937	Patent families; Public research-industry co-publications; State of cluster development†
Knowledge absorption	3	0.813	High-tech imports; ICT services imports; Intellectual property payments
Knowledge creation	3	0.907	PCT patents by inventor origin; Scientific and technical articles; Utility models by origin
Knowledge impact	2	0.777	Labor productivity growth; High-tech manufacturing

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Knowledge diffusion	3	0.871	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.811	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.739	Creative goods exports; National feature films
Online creativity	1	0.639	GitHub commits

Finding: At the same 42-indicator budget, Strategy 1a (pillar-level) achieves a rank correlation of 0.941, compared to 0.929 for Strategy 2a (sub-pillar-level). Strategy 1a was therefore used for the SIMPROF clustering comparison below.

Table 8: Rand Index: full clustering vs. proportional-subset (Strategy 1a) clustering

Comparison	Rand Index	Full Indicator Count	Subset Indicator Count	Strategy
Full_vs_ProportionalSubset	0.885	78	42	Strategy1a_ProportionalPillar

Finding: The 42-indicator proportional subset recovers a Rand Index of 0.885 - notably *lower* than Strategy 2’s 21-indicator Rand Index (0.902) despite covering double the indicators and scoring higher on rank correlation. Rand Index is not monotonic in subset size or in rank correlation: it depends on where SIMPROF’s significance testing happens to draw cluster boundaries, which can shift disproportionately with a different indicator mix. At the exact 42-indicator budget, a *random* subset already averages a rank correlation of 0.932 (Fig 2 above) - only marginally below Strategy 1a’s 0.941 - so exhaustive search is not clearly outperforming chance at this budget either, echoing the pattern seen at the smaller 21-indicator budget above.

4.2 Variable Perspective

4.2.1 PCA Analysis

The table below shows the number of principal components needed to capture specified levels of variance in the full 78-indicator dataset:

Table 9: Components needed for target variance levels (full PCA)

Variance Level	Components Needed
80%	17
90%	26
95%	36

Finding: The first 17 of 78 components already capture 80% of total variance, confirming substantial redundancy in the full indicator set - a necessary condition for indicator reduction to be viable at all. Note this is a property of the full PCA (unconstrained linear combinations of all 78 indicators), not directly comparable to how many *raw indicators* a subset needs - see the r_m vs. #PCs comparison below.

4.2.2 Reflective Indicators per Sub-Pillar (Strategy 2 - Headline Subset)

Per the project’s specified methodology, the headline recommended subset chooses one variable per sub-pillar (Level 2 aggregate), selecting within each sub-pillar the indicator with the best local r_m (“reflective indicator” - the one that best represents its sub-pillar’s own variance structure):

Table 10: Strategy 2: reflective indicator per sub-pillar

Sub-Pillar	r_m of Best Subset	Selected Indicators
Institutional environment	0.938	Operational stability for businesses*
Regulatory environment	0.975	Rule of law*
Business environment	0.834	Entrepreneurship policies and culture†
Education	0.626	School life expectancy
Tertiary education	0.615	Tertiary enrolment
Research and development (R&D)	0.793	Gross expenditure on R&D
Information and communication technologies (ICTs)	0.877	ICT access*
General infrastructure	0.647	Logistics performance*
Ecological sustainability	0.588	ISO 14001 environmental certificates
Credit	0.650	Finance for startups and scale-ups†
Investment	0.750	Joint venture deals

Sub-Pillar	r_m of Best Subset	Selected Indicators
Trade, diversification and market scale	0.639	Domestic industry diversification
Knowledge workers	0.802	Knowledge-intensive employment
Innovation linkages	0.727	University–industry R&D collaboration†
Knowledge absorption	0.508	Research talent
Knowledge creation	0.638	PCT patents by inventor origin
Knowledge impact	0.592	High-tech manufacturing
Knowledge diffusion	0.649	Production and export complexity
Intangible assets	0.613	Industrial designs by origin
Creative goods and services	0.550	Entertainment and media market
Online creativity	0.639	GitHub commits

Finding: This 21-indicator subset achieves $r_m = 0.826$, i.e. $r_m^2 = 68.2\%$ of the full dataset’s PCA variance structure.

4.2.3 Strategy 1 (Flat, 3 per Pillar) - Comparison

Table 11: Strategy 1: best 3 indicators per pillar (flat)

Pillar	r_m of Best Subset	Selected Indicators
Institutions	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	0.757	Expenditure on education; Gross expenditure on R&D; Tertiary enrolment
Infrastructure	0.784	Gross capital formation; ICT access*; Low-carbon energy use
Market sophistication	0.764	Domestic credit to private sector; Domestic market scale; Joint venture deals
Business sophistication	0.746	Females employed w/advanced degrees; High-tech imports; University–industry R&D collaboration†
Knowledge and technology outputs	0.702	Software spending; Production and export complexity; Patents by origin

Pillar	r_m of Best Subset	Selected Indicators
Creative outputs	0.697	Entertainment and media market; National feature films; Industrial designs by origin

Finding: Strategy 1 reaches $r_m = 0.854$ ($r_m^2 = 72.9\%$) - marginally *higher* than Strategy 2’s 0.826 at the same 21-indicator size. Strategy 2 is nonetheless used as the headline recommendation, not Strategy 1, because the project’s methodology specifically calls for “one variable per level-2 aggregate, justified by reflective indicator” rather than whichever flat strategy happens to score marginally higher - the sub-pillar-level selection is the specified deliverable, and the score difference between the two is small enough not to override that.

4.2.4 Strategies 1a and 2a (Proportional Allocation) - Comparison

Table 12: Strategy 1a: proportional allocation across pillars

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutions	3	0.965	Entrepreneurship policies and culture†; Policy stability for doing business†; Regulatory quality*
Human capital and research	6	0.898	Expenditure on education; Global corporate R&D investors; Researchers; Graduates in science and engineering; Tertiary enrolment; Tertiary inbound mobility
Infrastructure	5	0.900	GDP/unit of energy use; Gross capital formation; ICT access*; ISO 14001 environmental certificates; Low-carbon energy use
Market sophistication	6	0.926	Applied tariff rate, weighted avg.; Domestic industry diversification; Joint venture deals; Late VC deals intensity; Market capitalization; Loans from microfinance institutions
Business sophistication	8	0.932	Emerging knowledge workforce; FDI net inflows; GERD performed by business; High-tech imports; ICT services imports; Intellectual property payments; Patent families; State of cluster development†
Knowledge and technology outputs	8	0.915	Software spending; Labor productivity growth; High-tech exports; ICT services exports; ISO 9001 quality certificates; PCT patents by inventor origin; Unicorn valuation; Utility models by origin

Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Creative outputs	6	0.865	Mobile app creation; Intangible asset intensity, top 15; Creative goods exports; Entertainment and media market; National feature films; Trademarks by origin

Table 13: Strategy 2a: proportional allocation across sub-pillars

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Institutional environment	1	0.938	Operational stability for businesses*
Regulatory environment	1	0.975	Rule of law*
Business environment	1	0.834	Entrepreneurship policies and culture†
Education	3	0.882	Expenditure on education; PISA scales in reading, maths and science; Pupil-teacher ratio, secondary
Tertiary education	2	0.840	Graduates in science and engineering; Tertiary inbound mobility
Research and development (R&D)	2	0.922	Global corporate R&D investors; Researchers
Information and communication technologies (ICTs)	2	0.966	Government's online service; <i>ICT use</i>
General infrastructure	2	0.867	Electricity output; Gross capital formation
Ecological sustainability	1	0.588	ISO 14001 environmental certificates
Credit	1	0.650	Finance for startups and scale-ups†
Investment	3	0.980	Joint venture deals; Late VC deals intensity; Market capitalization
Trade, diversification and market scale	1	0.639	Domestic industry diversification
Knowledge workers	3	0.944	Females employed w/advanced degrees; GERD financed by business; GERD performed by business

Sub-Pillar	Indicators Allocated	r_m of Best Subset	Selected Indicators
Innovation linkages	3	0.937	Patent families; Public research–industry co-publications; State of cluster development†
Knowledge absorption	3	0.813	High-tech imports; ICT services imports; Intellectual property payments
Knowledge creation	3	0.907	PCT patents by inventor origin; Scientific and technical articles; Utility models by origin
Knowledge impact	2	0.777	Labor productivity growth; High-tech manufacturing
Knowledge diffusion	3	0.871	High-tech exports; Intellectual property receipts; ISO 9001 quality certificates
Intangible assets	2	0.811	Global brand value, top 5,000; Trademarks by origin
Creative goods and services	2	0.739	Creative goods exports; National feature films
Online creativity	1	0.639	GitHub commits

Finding: At 42 indicators, Strategy 1a reaches $r_m = 0.963$ ($r_m^2 = 92.6\%$) and Strategy 2a reaches $r_m = 0.946$ ($r_m^2 = 89.5\%$) - both a large step up from the 21-indicator strategies, as expected with double the budget.

4.2.5 Flat-Allocation Sweep: 1-4 Indicators per Pillar

To see how r_m grows as the flat per-pillar allocation increases, Strategy 1’s search was repeated for 1, 2, 3, and 4 indicators per pillar:

Table 14: r_m as the flat per-pillar allocation increases

Indicators per Pillar	r_m	Total Indicators
1	0.705	7
2	0.802	14
3	0.854	21
4	0.900	28

Finding: r_m rises from 0.705 at 7 variables (1/pillar) to 0.9 at 28 variables (4/pillar) - a steep gain per additional variable early on, consistent with the PCA finding above that a small

number of dimensions already captures most of the variance structure.

4.2.6 Random-Subset r_m Curve vs. #PCs Needed

For the same subset sizes used in the country perspective's Fig 2, r_m was computed for 20 random subsets per size. r_m is far more expensive per evaluation than the rank-correlation metric - it requires regressing all 78 principal components on the candidate subset - so fewer random draws were used per size here than Fig 2's 1,000.

Table 15: Random-subset r_m by size, vs. cumulative variance explained by the first k principal components

Subset Size (k)	r_m	Subset Variance Captured	PCA Variance (k components)
8	0.695	0.484	0.649
15	0.791	0.626	0.785
23	0.857	0.734	0.878
31	0.900	0.811	0.931
38	0.925	0.856	0.959
46	0.951	0.904	0.978
54	0.969	0.940	0.989
62	0.984	0.968	0.996
69	0.991	0.983	0.999

Finding: At $k = 21$ (interpolating between $k = 15$ and $k = 23$ above), a random subset averages roughly the same r_m as the exhaustive Strategies 1/2 report (0.854/0.826) - the same pattern seen in the country perspective, where exhaustive within-group search does not clearly beat chance at this budget. The PCA-variance column is consistently higher than a same-sized *indicator* subset's r_m^2 at every k - expected, since the first k principal components are unconstrained linear combinations of all 78 indicators, while a subset is restricted to k raw indicators; this gap is the practical cost of requiring the final subset to be actual, interpretable indicators rather than statistical composites.

4.2.7 Headline Recommended Subset

The 21-indicator subset from Strategy 2 above, with full indicator names:

Table 16: Headline subset: one reflective indicator per sub-pillar

Sub-Pillar	Indicator	r_m of Best Subset
Institutional environment	Operational stability for businesses*	0.938
Regulatory environment	Rule of law*	0.975
Business environment	Entrepreneurship policies and culture†	0.834
Education	School life expectancy	0.626
Tertiary education	Tertiary enrolment	0.615
Research and development (R&D)	Gross expenditure on R&D	0.793
Information and communication technologies (ICTs)	ICT access*	0.877
General infrastructure	Logistics performance*	0.647
Ecological sustainability	ISO 14001 environmental certificates	0.588
Credit	Finance for startups and scale-ups†	0.650
Investment	Joint venture deals	0.750
Trade, diversification and market scale	Domestic industry diversification	0.639
Knowledge workers	Knowledge-intensive employment	0.802
Innovation linkages	University–industry R&D collaboration†	0.727
Knowledge absorption	Research talent	0.508
Knowledge creation	PCT patents by inventor origin	0.638
Knowledge impact	High-tech manufacturing	0.592
Knowledge diffusion	Production and export complexity	0.649
Intangible assets	Industrial designs by origin	0.613
Creative goods and services	Entertainment and media market	0.550
Online creativity	GitHub commits	0.639

Finding: This subset captures r_m^2 68.2% of the full dataset’s PCA variance structure and recovers a Rand Index of 0.902 against the full-data country clustering, while cutting the indicator count from 78 to 21 (a 73% reduction).

5 Cross-Strategy Comparison

Table 17: Country perspective: all four strategies compared

Strategy	Indicators	Rank Correlation
1 (flat, 3/pillar)	21	0.834
2 (flat, 1/sub-pillar)	21	0.841
1a (proportional, pillar)	42	0.941
2a (proportional, sub-pillar)	42	0.929

Table 18: Variable perspective: all four strategies compared

Strategy	Indicators	r_m	r_m^2 (%)
1 (flat, 3/pillar)	21	0.854	72.9
2 (flat, 1/sub-pillar, headline)	21	0.826	68.2
1a (proportional, pillar)	42	0.963	92.6
2a (proportional, sub-pillar)	42	0.946	89.5

Finding: Across both perspectives and both metrics, proportional allocation at 42 indicators outperforms flat allocation at 21 indicators - but budget size, not the allocation rule itself, explains most of that gap (see the random-baseline comparisons above, including the exact 42-indicator random baseline). Within a fixed budget, the two allocation rules perform comparably; the practical case for proportional allocation is representational (no pillar/sub-pillar is systematically over- or under-weighted relative to its actual size in the index), not a demonstrated statistical edge over flat allocation on these metrics alone.

6 Conclusions

1. **Meaningful reduction is possible, with a clear methodological trade-off.** The headline 21-indicator subset (one reflective indicator per sub-pillar) captures r_m^2 68.2% of the full PCA’s variance structure and a Rand Index of 0.902 against the full clustering, at a 73% reduction in indicator count.
2. **Exhaustive within-group search does not clearly outperform random selection at the 21-indicator budget,** for either metric: both the country-perspective and variable-perspective exhaustive strategies score close to what an unconstrained random subset of similar size averages. The gain from exhaustive search becomes more visible at the larger 42-indicator budget, though even there the margin over a random baseline of the same size is modest.

3. **Flat and proportional allocation perform comparably at a fixed budget;** proportional allocation’s main practical benefit is representational (no pillar/sub-pillar is systematically over- or under-weighted relative to its actual size in the index), not a demonstrated statistical edge over flat allocation at matched size.
4. **Rand Index is not monotonic with rank correlation or subset size** - the 42-indicator proportional subset scores a lower Rand Index than the 21-indicator flat subset despite a higher rank correlation, because SIMPROF’s significance-tested cluster boundaries can shift disproportionately with a different indicator mix.
5. **The r_m criterion treats “information” as PCA variance,** which may downweight indicators with low variance but high substantive/policy relevance; the headline subset should be reviewed against domain expertise, not adopted on statistical grounds alone.

7 Limitations

1. **Single-year analysis.** This analysis covers GII 2025 only; temporal stability of the recommended subset across years has not been assessed.
2. **PCA-defined “information.”** The r_m criterion equates information content with PCA variance structure; a substantively important but low-variance indicator could be excluded even if policy-relevant.
3. **Local search proxy.** Exhaustive search within each pillar/sub-pillar uses a local r_m computed on that group’s own sub-PCA (not the full 78-indicator PCA) as a tractable stand-in for an otherwise infeasible full search; the assembled subset is re-scored against the true full-dataset metric for reporting, but the *search* itself never directly optimizes that true metric.
4. **Linear relationships assumed.** Both the rank-correlation and r_m criteria are built on linear/monotonic structure (PCA, Euclidean distance); non-linear relationships between indicators are not captured.
5. **Domain-specific validity not tested.** Whether the recommended subset remains meaningful to GII’s intended policy audience has not been evaluated here and would require domain-expert review.

8 Data Availability

All data and code are available in this repository:

- Raw data: `gii-data/idata.xlsx`, `gii-data/imeta.xlsx`
- Processed data: `data/gii_data_standardized.csv`, `data/gii_data_complete_cases.csv`
- Analysis scripts: `scripts/00_setup.R` through `scripts/04_variable_perspective.R`, shared functions in `scripts/R/`

- Results: `outputs/` (CSVs for tabular results, `.rds` for clustering objects)

9 References

- Algursan, A.S., M. Durand, H. Fu, et al. 2026. “Counting What Counts: A Compass of Progress for People and Planet.” In *Report of the Secretary-General's Independent High-Level Expert Group on Beyond GDP*. United Nations.
- Benczur, P., A. Boskovic, J. Cariboni, et al. 2024. *Sustainable and Inclusive Wellbeing: The Road Forward*. Publications Office of the European Union.
- Boskovic, Ana, Enrico Giovannini, Andrea Pagano, Alina-Mihaela Sandor, and Peter Benczur. 2024. *Measuring Sustainable and Inclusive Wellbeing: A Multidimensional Dashboard Approach*. Publications Office of the European Union.
- Cadima, Jorge F.C.L., and Ian T. Jolliffe. 2001. “Variable Selection and the Interpretation of Principal Subspaces.” *Journal of Agricultural, Biological, and Environmental Statistics* 6 (1): 62–79.
- Clarke, K.R., P.J. Somerfield, and R.N. Gorley. 2008. “Testing of Null Hypothesis in Exploratory Community Analyses: Similarity Profiles and Biota-Environment Linkage.” *Journal of Experimental Marine Biology and Ecology* 366: 56–69.
- McCabe, George P. 1984. “Principal Variables.” *Technometrics* 26 (2): 137–44.
- (WIPO), World Intellectual Property Organization. 2024. *Global Innovation Index 2025: Innovation at a Crossroads*. WIPO. <https://doi.org/10.34667/tind.58864>.

10 Appendix

11 Data Imputation Documentation

Missing data is an inherent challenge in composite index construction. This analysis used k-Nearest Neighbors (kNN) imputation to address missing values before conducting the main analyses.

11.1 Imputation Process

- **Method:** k-Nearest Neighbors (kNN) imputation

- **Parameters:** $k = 5$ neighbors (VIM::kNN, other parameters left at package defaults)
- **Implementation:** R VIM package
- **Indicators Used:** Level 1 indicators from IMETA metadata

11.2 Missing Data Statistics

Table 19: Imputation Process Summary

Metric	Count
Total Missing Values	1211
Variables with Missing Data	58
Countries with Missing Data	135

Note: 1211 missing values across 58 of 78 indicators, affecting 135 of 139 countries, were imputed prior to standardization.

12 Descriptive Statistics Tables

12.1 Descriptive Statistics - Original Data

The table below presents descriptive statistics for all Level 1 indicators in the original dataset before imputation:

Table 20: Descriptive Statistics - Original Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Government effectiveness*	0.15	0.03	-1.75	2.32	-0.71	0.29
Operational stability for businesses*	2.23	2.15	0.70	4.45	0.03	0.58
Regulatory quality*	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law*	0.07	-0.12	-2.15	1.97	-0.77	0.29
Entrepreneurship policies and culture†	4.72	4.72	2.92	7.25	-0.21	0.45
Policy stability for doing business†	4.03	3.98	1.83	6.45	-0.69	0.02
Expenditure on education	4.25	4.16	0.35	9.39	0.80	0.29
Government funding/pupil, secondary	19.60	19.41	6.00	51.06	1.57	0.74

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
PISA scales in reading, maths and science	439.44	439.29	337.45	579.03	-0.84	0.07
Pupil-teacher ratio, secondary	15.89	13.53	6.14	68.13	9.84	2.46
School life expectancy	14.17	14.36	6.34	20.85	-0.23	-0.28
Graduates in science and engineering	23.29	23.14	9.56	41.10	-0.33	0.17
Tertiary enrolment	51.51	54.40	2.71	166.67	-0.02	0.37
Tertiary inbound mobility	7.34	4.24	0.00	69.73	15.40	3.35
QS university ranking*	21.29	10.43	0.00	97.63	0.29	1.14
Global corporate R&D investors	926.35	0.00	0.00	36142.79	60.93	7.11
Gross expenditure on R&D	1.01	0.61	0.01	6.35	4.50	1.92
Researchers	2269.70	922.47	4.50	10413.00	0.53	1.21
Government's online service*	0.68	0.72	0.17	1.00	-0.95	-0.40
ICT access*	83.02	88.25	24.33	100.00	0.54	-1.12
ICT use*	79.65	84.47	29.84	99.36	1.78	-1.49
Electricity output	4936.33	3183.45	29.76	53263.16	24.28	4.06
Gross capital formation	24.49	23.70	11.60	44.56	0.63	0.74
Logistics performance*	3.12	3.10	2.10	4.30	-1.10	0.19
GDP/unit of energy use	12.23	11.06	2.16	41.21	4.01	1.59
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
Domestic credit to private sector	60.33	48.53	8.14	248.76	2.51	1.51
Finance for startups and scale-ups†	4.49	4.49	1.99	6.90	-0.48	0.09
Loans from microfinance institutions	2.09	1.05	0.00	17.34	12.83	3.33
Joint venture deals	0.21	0.06	0.00	4.55	56.21	6.66
Late VC deals intensity	0.32	0.03	0.00	11.43	55.80	7.13
Market capitalization	80.85	37.45	0.20	1507.77	55.74	6.96
Venture capital investors	0.65	0.14	0.01	29.23	101.26	9.79
Venture capital recipients	0.24	0.08	0.01	5.50	67.12	7.42
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Domestic industry diversification	0.18	0.14	0.07	0.55	3.07	1.61
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Females employed w/advanced degrees	13.49	12.78	0.14	31.04	-1.14	0.26
GERD financed by business	34.31	38.44	0.00	80.84	-1.07	-0.07
GERD performed by business	0.77	0.38	0.00	5.90	7.67	2.33

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Knowledge-intensive employment	27.07	23.94	2.02	68.01	-0.95	0.31
Patent families	1.30	0.03	0.00	73.67	114.09	10.30
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
State of cluster development†	4.00	3.90	2.59	5.40	-0.79	0.24
WU Rankings industry and global outlook	55.74	49.40	26.70	97.10	-0.99	0.58
University–industry R&D collaboration†	3.97	3.86	2.30	6.72	-0.12	0.38
FDI net inflows	3.93	2.50	- 161.34	224.25	66.45	3.27
High-tech imports	9.33	8.01	1.30	55.19	21.43	3.65
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
Research talent	33.25	32.27	0.30	81.91	-1.19	0.21
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Patents by origin	2.54	0.62	0.00	61.56	39.40	5.92
PCT patents by inventor origin	0.62	0.07	0.00	7.57	11.91	3.30
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Utility models by origin	1.55	0.16	0.00	88.51	84.62	9.17
Software spending	0.24	0.21	0.00	1.26	3.08	1.42
Labor productivity growth	0.83	0.86	-4.57	6.26	0.84	-0.01
High-tech manufacturing	25.19	21.99	1.04	79.65	0.15	0.75
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Production and export complexity	0.09	-0.05	-2.18	2.29	-0.52	0.23
High-tech exports	4.57	1.51	0.02	48.39	12.41	3.08
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Global brand value, top 5,000	3.59	0.78	0.00	21.12	1.78	1.64
Intangible asset intensity, top 15	41.66	56.72	- 371.30	89.79	28.96	-4.81
Industrial designs by origin	2.21	0.92	0.00	23.28	11.87	3.25
Trademarks by origin	36.72	28.95	0.00	202.33	7.36	2.24
Creative goods exports	1.27	0.38	0.00	11.50	7.00	2.60
Cultural and creative services exports	0.96	0.46	0.00	18.31	53.87	6.48
Entertainment and media market	0.93	0.47	0.02	3.59	-0.06	0.92

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
National feature films	5.01	2.67	0.00	46.93	18.95	3.70
Mobile app creation	1379627.73	181351.44	0.04	98744342.48	123.62	10.98
GitHub commits	1326116.27	410927.87	2571.55	14402260.97	14.47	2.99
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93

12.2 Descriptive Statistics - Complete Data

The table below presents descriptive statistics for all Level 1 indicators in the complete dataset after kNN imputation and case-wise cleaning:

Table 21: Descriptive Statistics - Complete Data (n=78 indicators)

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
Operational stability for businesses*	2.23	2.15	0.70	4.45	0.03	0.58
Government effectiveness*	0.15	0.03	-1.75	2.32	-0.71	0.29
Regulatory quality*	0.16	0.11	-2.03	2.31	-0.72	0.21
Rule of law*	0.07	-0.12	-2.15	1.97	-0.77	0.29
Policy stability for doing business†	4.03	3.98	1.83	6.45	-0.65	0.02
Entrepreneurship policies and culture†	4.63	4.54	2.92	7.25	0.28	0.64
Expenditure on education	4.24	4.14	0.35	9.39	0.83	0.31
Government funding/pupil, secondary	18.48	17.96	6.00	51.06	1.96	0.85
School life expectancy	13.92	13.94	6.34	20.85	-0.42	-0.13
PISA scales in reading, maths and science	419.11	403.60	337.45	579.03	-0.43	0.70
Pupil-teacher ratio, secondary	15.80	13.28	6.14	68.13	10.01	2.48
Tertiary enrolment	50.58	53.42	2.71	166.67	0.00	0.41
Graduates in science and engineering	22.88	22.88	9.56	41.10	-0.31	0.25
Tertiary inbound mobility	6.61	3.14	0.00	69.73	17.76	3.58
Researchers	1922.18	669.77	4.50	10413.00	1.08	1.44
Gross expenditure on R&D	0.88	0.42	0.01	6.35	5.71	2.16
Global corporate R&D investors	933.55	0.00	0.00	36142.79	61.25	7.12
QS university ranking*	21.29	10.43	0.00	97.63	0.29	1.14
ICT access*	82.88	88.19	24.33	100.00	0.47	-1.09

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
ICT use*	78.81	83.81	29.84	99.36	1.28	-1.36
Government's online service*	0.68	0.73	0.17	1.00	-0.94	-0.41
Electricity output	4663.61	2879.19	29.76	53263.16	25.59	4.17
Logistics performance*	3.04	2.90	2.10	4.30	-0.96	0.45
Gross capital formation	24.48	23.70	11.60	44.56	0.67	0.75
GDP/unit of energy use	12.00	10.93	2.16	41.21	4.46	1.69
Low-carbon energy use	22.00	18.38	0.01	82.64	0.82	1.07
ISO 14001 environmental certificates	2.05	0.99	0.04	11.87	2.95	1.82
Finance for startups and scale-ups†	4.25	4.11	1.99	6.90	-0.45	0.48
Domestic credit to private sector	60.08	48.30	8.14	248.76	2.47	1.51
Loans from microfinance institutions	1.43	0.63	0.00	17.34	26.25	4.59
Market capitalization	58.57	25.34	0.20	1507.77	88.80	8.73
Venture capital investors	0.56	0.11	0.01	29.23	120.91	10.68
Venture capital recipients	0.22	0.07	0.01	5.50	73.92	7.78
Late VC deals intensity	0.25	0.01	0.00	11.43	71.99	8.09
Joint venture deals	0.18	0.04	0.00	4.55	66.13	7.21
Applied tariff rate, weighted avg.	4.16	2.34	0.00	47.54	40.98	5.09
Domestic industry diversification	0.19	0.17	0.07	0.55	2.05	1.19
Domestic market scale	1370.77	253.12	4.14	37072.09	47.47	6.55
Emerging knowledge workforce	39.00	35.41	17.63	66.87	-0.99	0.53
Knowledge-intensive employment	26.67	23.31	2.02	68.01	-0.89	0.38
Females employed w/advanced degrees	13.06	11.20	0.14	31.04	-1.04	0.35
GERD performed by business	0.50	0.06	0.00	5.90	11.94	2.97
GERD financed by business	28.45	28.73	0.00	80.84	-1.02	0.34
Public research–industry co-publications	2.01	1.34	0.22	9.07	2.97	1.78
University–industry R&D collaboration†	3.95	3.83	2.30	6.72	-0.05	0.42
WU Rankings industry and global outlook	51.65	42.20	26.70	97.10	-0.25	1.02
State of cluster development†	3.99	3.84	2.59	5.40	-0.75	0.28
Patent families	1.29	0.03	0.00	73.67	114.91	10.33
Intellectual property payments	1.05	0.54	0.00	23.25	64.28	7.17
High-tech imports	9.32	8.03	1.30	55.19	21.61	3.67
ICT services imports	1.75	1.44	0.01	16.01	31.07	4.44

Indicator	Mean	Median	Min	Max	Kurto- sis	Skew- ness
FDI net inflows	3.91	2.49	- 161.34	224.25	66.93	3.29
Research talent	23.49	11.34	0.30	81.91	-0.61	0.86
Patents by origin	2.51	0.61	0.00	61.56	39.97	5.96
PCT patents by inventor origin	0.58	0.06	0.00	7.57	13.22	3.45
Utility models by origin	1.05	0.16	0.00	88.51	136.59	11.64
Scientific and technical articles	13.42	9.94	0.55	47.72	0.52	1.13
Citable documents H-index	466.22	281.00	53.00	3213.00	7.98	2.39
Labor productivity growth	0.83	0.86	-4.57	6.26	0.87	0.00
Unicorn valuation	1.09	0.00	0.00	21.85	27.52	4.75
Software spending	0.24	0.20	0.00	1.26	3.09	1.42
High-tech manufacturing	22.20	18.35	1.04	79.65	0.37	0.94
Intellectual property receipts	0.48	0.07	0.00	5.36	9.15	3.01
Production and export complexity	0.08	-0.06	-2.18	2.29	-0.43	0.26
High-tech exports	4.53	1.51	0.02	48.39	12.49	3.09
ICT services exports	2.99	1.68	0.00	36.98	26.94	4.55
ISO 9001 quality certificates	5.46	3.23	0.04	29.87	3.42	1.80
Intangible asset intensity, top 15	32.37	39.35	- 371.30	89.79	34.44	-4.68
Trademarks by origin	36.74	28.99	0.00	202.33	7.57	2.26
Global brand value, top 5,000	3.20	0.47	0.00	21.12	2.47	1.81
Industrial designs by origin	2.12	0.84	0.00	23.28	12.62	3.34
Cultural and creative services exports	0.86	0.42	0.00	18.31	60.72	6.87
National feature films	4.47	2.69	0.00	46.93	26.15	4.26
Entertainment and media market	0.54	0.17	0.02	3.59	2.74	1.84
Creative goods exports	1.27	0.38	0.00	11.50	7.07	2.61
Top-level domains (TLDs)	117.16	22.31	0.12	2538.57	46.17	5.93
GitHub commits	1357449.63	411718.75	2571.55	14402260.97	13.92	2.91
Mobile app creation	1305696.24	155386.88	0.04	98744342.48	131.12	11.31